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[MANDATORY NOTICE]

SYNTHETIC DEMONSTRATION DATA

This document is fictional test data created for the VYASA prototype.

It is not an actual operating procedure, engineering standard,
regulatory requirement, or safety instruction.

All numerical values are example synthetic demonstration values only.

SECTION 1: SCOPE, EQUIPMENT & OPERATING PARAMETERS

1.1 Scope & Equipment Covered

This synthetic procedure covers operational parameters and condition monitoring for centrifugal process pumps in boiler feed and hydrocarbon transfer service, specifically example equipment unit pump P-101 and pump P-102.

1.2 Primary Operating Parameters

- Rated volumetric flow capacity: 120 m³/h at rated RPM.
- Example synthetic demonstration suction pressure: 2.5 bar gauge.
- Example synthetic demonstration discharge pressure: 14.2 bar gauge.
- Mechanical seal flush configuration: API Plan 11 with cooling jacket.

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SECTION 2: PUMP P-102 VIBRATION & TEMPERATURE THRESHOLDS

2.1 Vibration Monitoring Limits for Pump P-102

Condition monitoring sensors measure overall velocity RMS on drive-end and non-drive-end bearings.

Example / Synthetic Demonstration Values for Pump P-102:

- Normal baseline vibration: < 2.8 mm/s RMS.
- Example synthetic demonstration vibration warning threshold for pump P-102: vibration > 5.0 mm/s.
- Example synthetic demonstration critical vibration trip threshold for pump P-102: vibration > 7.5 mm/s.

2.2 Bearing Temperature Observations for Pump P-102

- Normal operating bearing temperature: 60.0 °C to 75.0 °C.
- Example synthetic demonstration bearing temperature warning threshold: > 80.0 °C.
- Example synthetic demonstration bearing temperature critical trip threshold: > 95.0 °C.

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SECTION 3: STARTUP, SHUTDOWN OBSERVATIONS & MAINTENANCE INDICATORS

3.1 Pre-Startup Verification Checklist

- Confirm seal flush flow and vent casing to eliminate trapped air or vapor.
- Check lubricating oil level in bearing housing sight glass (must be at mid-level 50%).
- Rotate pump shaft manually to verify free rotation without binding.

3.2 Abnormal Operating Observations & Maintenance Indicators

- Cavitation noise: distinctive rattling or gravel-sound in casing indicates insufficient NPSH available.
- Mechanical seal leakage: example synthetic observation limit of > 5 drops/minute indicates seal wear requiring scheduled overhaul.
- Routine maintenance interval: example synthetic observation inspection every 2,000 running hours.